

MTP GAPS Implementation

Fixing two problems in "Long-term Fairness in Ride-Hailing Platform"

The paper has two main formulas.

Formula 1 — value of giving a trip to a driver:

$$U = \text{Geo}(d, s) - \text{Geo}(s, g)$$

Formula 2 — how the paper measures fairness:

$$\text{Fairness} = \text{Var}(\text{total weekly earning of each driver})$$

1. The two problems I found

Problem 1 (Gap 1): the formulas ignore traffic and ignore working hours

Part A — traffic is ignored. In Formula 1 there is only distance, no time. The paper also says clearly in **Section 5.1** that it takes the average travel time for each route and uses that one number for the whole day. So a 5 km trip at 4 in the morning and the same 5 km trip at 6 in the evening are treated as exactly equal. **In real Manhattan traffic they are not:**

Same 5 km trip	Actual time taken
At 4 AM (empty roads)	about 12 minutes
At 6 PM (peak traffic)	about 26 minutes

I checked this on the paper's own data. For the same start and end point, the travel time at peak is **2.11 times the off-peak time on average**, and up to **4.35 times in the worst case**. **So the paper is throwing away a real effect.**

Part B — working hours are ignored. Formula 2 only looks at total earnings. It never asks how long the driver was working. Take this example:

Driver	Earned in the week	Hours worked	Actual rate per hour
A (full-time)	Rs 6,000	60 hours	Rs 100 per hour
B (part-time)	Rs 6,000	15 hours	Rs 400 per hour

Both earned Rs 6,000, so variance = 0 and the paper calls this perfectly fair. But A is actually earning 4 times less per hour than B. The paper cannot see this at all.

Problem 2 (Gap 2): it never checks if an available driver got any work

The paper only looks at final earnings. It never checks whether a driver who was online actually received trips.

Driver	Hours online	Trips given	Sitting idle	Earned
X	8 hours	2	6 hours	Rs 5,000
Y	8 hours	10	1 hour	Rs 5,000

X got only 2 trips but they happened to be long ones, so earnings are equal and the paper says fair. But X wasted 6 hours sitting with the app open and getting nothing. That is a real unfairness which is nowhere in the paper.

2. What I did with the data

The raw file is 1.9 GB with 1,22,10,952 trip records. Each row is one taxi trip with pickup time, drop time, pickup and drop location, distance and fare. One important thing: the file has no driver ID. It only records trips, not which driver did them. This matters a lot, and I will explain it below.

#	What I did	Before to After	Why it was needed
1	Cleaned the data	1,22,10,952 to 1,03,00,738 trips	Removed trips outside Manhattan, trips of 0 km, one trip showing 50,00,000 km (GPS error), negative fares, and trips shorter than 1 minute or longer than 2 hours
2	Divided the city into 87 areas	GPS points to 87 nodes	The model needs a fixed set of locations, not crores of GPS points. I tested several grid sizes and picked around 1.1 km because smaller areas did not have enough trips to calculate travel time properly
3	Made a distance table	87 x 87 table	The paper uses Geo(a,b) but never says how it got it. I took the median actual distance from real trips. Check: my road distance comes out 1.32 times the straight-line distance, and the known value for New York is about 1.3, so the table is correct
4	Made a travel-time table	1 number to 48 numbers per route	This is the fix for Problem 1 Part A. 48 = weekday/weekend times 24 hours. This is where the 2.11 times traffic difference shows up
5	Created 200 drivers	no drivers to 200 drivers with shifts	Since the data has no driver ID, drivers have to be created. 76 full-time (around 46 hours a week) and 124 part-time (around 14 hours a week). Without hours, Problem 1 Part B and Problem 2 cannot even be measured
6	Decision every 5 minutes instead of 1 hour	1 hour to 5 minutes	An average Manhattan trip is 11 minutes. With 1-hour steps you cannot say whether a driver was busy or free, and that is exactly what Problem 2 needs
7	Used the full day instead of 2 peak hours	2 hours a day to 24 hours a day	The paper only uses 2 hours of each day. That means a driver can work at most 14 hours a week, so a 40-hour full-time driver is impossible. I ran the paper's own setting to confirm: 0 out of 200 drivers could reach full-time

Important point: the fully cleaned data (1,03,00,738 trips) is used to build everything — the 87 areas, the distance table, the travel-time table and the prediction model. Only the final allocation experiment runs on a smaller sample of requests, and the paper does the same thing (it uses a 5% sample).

3. The new formulas I made

Fix for traffic (Gap 1, Part A)

First I made a traffic number:

$$c = (\text{travel time on this route at this hour}) / (\text{average travel time on this route})$$

Value of c	Meaning
c = 1	Normal speed for that route
c = 2	Taking twice as long as usual, heavy traffic
c = 0.5	Taking half the usual time, empty roads

Then I put c into the utility formula:

$$U_{\text{new}} = \text{Geo}(d, s) / c(s, d) - \text{Geo}(s, g) \times c(g, s)$$

1. The paid part is DIVIDED by c. If a trip takes twice as long, the driver is stuck in traffic for double the time for the same money, so it is worth less to him.
2. The empty-travel part is MULTIPLIED by c. Driving empty in traffic is worse than driving empty on a clear road: same distance, more time wasted, still no money.
3. **Very important check: if c = 1 everywhere, this formula becomes exactly the paper's original formula. I verified this in code and the difference came out exactly 0. So the paper is just a special case of my formula, which makes the comparison fair.**

Example

An 8 km trip and the passenger 5 km away:

Case	c	Utility
Paper (same at any time)	-	8 - 5 = +3 km (looks profitable)
Mine, 4 AM, clear roads	0.6	8/0.6 - 5(0.6) = +10.3 km
Mine, 6 PM, heavy traffic	2.0	8/2.0 - 5(2.0) = -6.0 km (actually a loss)

So the paper says this trip is good (+3) when it is actually bad (-6). I checked this over the whole data: the paper gets the good/bad sign wrong in 10.2% of cases, which is 1 out of every 10.

Fix for working hours (Gap 1, Part B)

$$\text{rate} = \text{total earning of the driver} / \text{hours he was online}$$

Then instead of measuring variance of total earning, I measure variance of this rate. I also split it into two parts using a standard result from probability called the law of total variance:

$$\text{Var}(\text{rate}) = \text{within-group part} + \text{between-group part}$$

Term	What it means in simple words
within-group	How unequal drivers are compared to drivers of their own type (one full-time driver vs another full-time driver)
between-group	How unequal the two groups are on average (full-time drivers as a group vs part-time drivers as a group). This is the real pay gap between the two types.
Groups used	Full-time = 40 or more hours a week. Part-time = 20 hours or less.

This split is not something I invented. It is a known mathematical identity, so within + between always adds up to the total exactly.

Fix for idle drivers (Gap 2)

$$\text{utilisation} = \text{time the driver was busy} / \text{time the driver was online}$$

So if a driver is online 8 hours and busy 2 hours, utilisation = 0.25. Then I measure variance of utilisation across drivers. If it is high, some drivers are getting work and others are sitting idle.

But there is one problem with this. If a driver himself chooses to be online at 4 AM when there are no passengers, his utilisation will be low, and that is not the app's fault. So I also made a second version:

$$\text{adjusted utilisation} = \text{driver's utilisation} / \text{what other drivers online at the same time got}$$

If this comes to 1, the driver did as well as everyone else working those same hours. If it is less than 1, the app really did give him less work. I report both numbers so nothing is hidden.

The final objective I optimise

Score = total earnings

- $w_1 \times (\text{within-group unfairness in rate})$
- $w_2 \times (\text{between-group unfairness in rate})$
- $w_3 \times (\text{unfairness in utilisation})$

w_1, w_2, w_3 are weights that decide how much importance to give to each part. One thing I found here: the paper fixes a scaling value $\omega = 0.6$ and does not say how it got it. When I used a fixed value, my utilisation term came out around 0.009 while a normal trip value is about 2.4. So that term was almost 300 times too small and was doing nothing at all. I fixed this by measuring the correct scale from an actual run. After that all my ablation results started behaving properly. This itself is a finding about the paper.

I also had to add one more thing. The paper's model only knows the driver's location. If the model does not know how far behind a driver is in earnings, it cannot correct it. So I added the driver's shortfall in rate and in utilisation into the model's state.

4. Results

All six methods were run on exactly the same setup: same 200 drivers, same 6,604 requests, same distance and travel-time tables. So any difference is only because of the allocation method. Utility is in km.

4.1 First, what each result column means

Column	Meaning	Good direction
Total Utility	Total earnings of all 200 drivers for the week	Higher
Var(total)	The paper's fairness measure. Spread of total weekly earnings.	Lower
Var(rate)	My fairness measure. Spread of earning per hour.	Lower
within	Part of Var(rate) coming from differences inside the same group	Lower
between	Part of Var(rate) coming from the gap between full-time and part-time drivers	Lower
FT / hr	Average earning per hour of full-time drivers	-
PT / hr	Average earning per hour of part-time drivers	-
ratio	PT rate divided by FT rate. 1.00 means both groups earn the same per hour, which is what we want.	Close to 1.00
Var(util)	Spread of how busy the drivers were kept	Lower
Var(util) adj	Same but after removing the effect of which hours the driver chose to work	Lower
idle	Number of drivers who got no trip at all in the whole week	0

4.2 Main result table

Method	Total Utility	Var(total)	Var(rate)	between	FT /hr	PT /hr	ratio	Var(util)	adj	idle
Greedy	11,708	701	1.809	0.5297	1.78	3.28	1.84	0.0202	0.426	1
REASSIGN	11,437	869	1.439	0.3060	1.83	2.97	1.62	0.0214	0.401	2
LAF	11,704	635	1.641	0.5973	1.75	3.34	1.91	0.0216	0.439	1
Balance Ride-Pooling	10,373	671	1.238	0.2516	1.66	2.69	1.62	0.0140	0.275	0
Kang et al. (reproduced)	10,221	657	1.251	0.2473	1.64	2.67	1.62	0.0129	0.253	1
Ours (Gap 1+2)	10,145	1,639	0.446	0.0011	1.93	1.86	0.97	0.0081	0.155	0

Main thing to notice in the 'ratio' column: all five earlier methods are between 1.62 and 1.91. That means part-time drivers are earning 62% to 91% more per hour than full-time drivers in every single one of them. This is not random, it happens because all of them are trying to equalise total earnings. **My method brings it to 0.97, which is almost exactly equal.**

4.3 My method compared to the paper's method

What is measured	Paper	Mine	Change
Total earnings	10,221	10,145	-0.7% (small loss)
Unfairness in earning per hour	1.251	0.446	-64.4% better
Gap between the two driver groups	0.2473	0.0011	-99.6% better
Full-time earning per hour	1.64	1.93	+17.4% (went up)
Part-time earning per hour	2.67	1.86	-30.2% (came down to equal)
PT / FT ratio (1.00 is equal)	1.62	0.97	almost equal now

Unfairness in busy time	0.0129	0.0081	-37.7% better
Same, after adjusting for shift choice	0.2527	0.1550	-38.6% better
Drivers with no work all week	1	0	removed completely

So by giving up only 0.7% of total earnings, the per-hour pay of the two driver groups became almost equal, and no driver was left without work.

4.4 One more finding: the paper's fairness formula actually creates the problem

I ran the same method with the fairness part switched off and switched on, to see what the paper's fairness formula is really doing.

Setting	FT per hour	PT per hour	Gap between groups
Fairness OFF (only earnings)	2.28	2.42	0.004 (almost no gap)
Paper's fairness ON	1.89	3.13	0.361 (90 times worse)

This is the important point. The paper's fairness formula does not just fail to see the group gap, it actually creates it. The reason is simple: if driver A works 50 hours and driver B works 10 hours, the only way to make their total earnings equal is to take work away from A and give it to B. So B's earning per hour shoots up. It is unavoidable with that formula.

4.5 How results improved as I added each fix

Step	Total Utility	Var(rate)	between	ratio	Var(util)	idle
No fairness at all	11,365	1.284	0.0062	1.08	0.0147	4
Paper's fairness added	10,221	1.251	0.2473	1.62	0.0129	1
My Gap 1 fixes added	10,191	0.426	0.0005	0.98	0.0123	0
My Gap 2 fix added (final)	10,145	0.446	0.0011	0.97	0.0081	0

Step 1 to 2: paper's fairness makes the group gap much worse (0.0062 to 0.2473) and the ratio jumps from 1.08 to 1.62.

Step 2 to 3: my Gap 1 fix brings Var(rate) down from 1.251 to 0.426 and the group gap almost to zero. Idle drivers also become 0.

Step 3 to 4: my Gap 2 fix brings Var(util) down from 0.0123 to 0.0081, about 34% better. This proves the Gap 2 part is doing real work and not just riding on Gap 1.

4.6 Checks I did to make sure it is not luck

Check	Result
Did it work in rupees/dollars also, not just km?	Yes. Paper: full-time \$6.62/hr vs part-time \$12.35/hr. Mine: \$8.20 vs \$7.93. So the gap is real, not a unit problem.
Did it work with different random settings?	Yes, ran 3 different random seeds. Paper's ratio stayed 1.84 every time (so the problem is systematic, not luck). Mine stayed 0.97.
Am I just trading earnings for fairness?	No. I tested 20 different weight settings. In 12 of them my method beat the paper on all six measures at the same time, including total earnings.
Is the prediction model working?	Yes. Demand prediction error 6.46 against a simple baseline of 8.86, so about 27% better. Traffic prediction error is 0.08, which is small.